CausalCache: Conditional High-Fidelity Restoration for Long-Horizon GUI Agents

Jiaxuan Luo, Zhanfeng Liao, Jiayao Teng, Yuan Wang

Paper version: arXiv 2608.22577v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Long-horizon GUI agents can retain complete action histories as compact text, but only a few historical screenshots fit in active context.*

Which past events regain their screenshots under a **fixed budget of B images**, while every event stays as text?

Motivation: three open gaps

1. **Recency by default.** Recent-B puts all B images on the latest events.
2. **Context matters.** A distant screenshot may help, or be redundant.
3. **Frozen policy:** weak, unreliable selectivity.

Evaluation setup

OSWorld-Verified · 361 | MobileWorld · 117 | GUI-Owl-1.5-8B frozen | Trained on desktop only | B = 4 images | MobileWorld: 3 runs | OSWorld: 1 run

- **Online:** closed-loop success on OSWorld and MobileWorld.
- **Held out:** MobileWorld never touches training or selection.

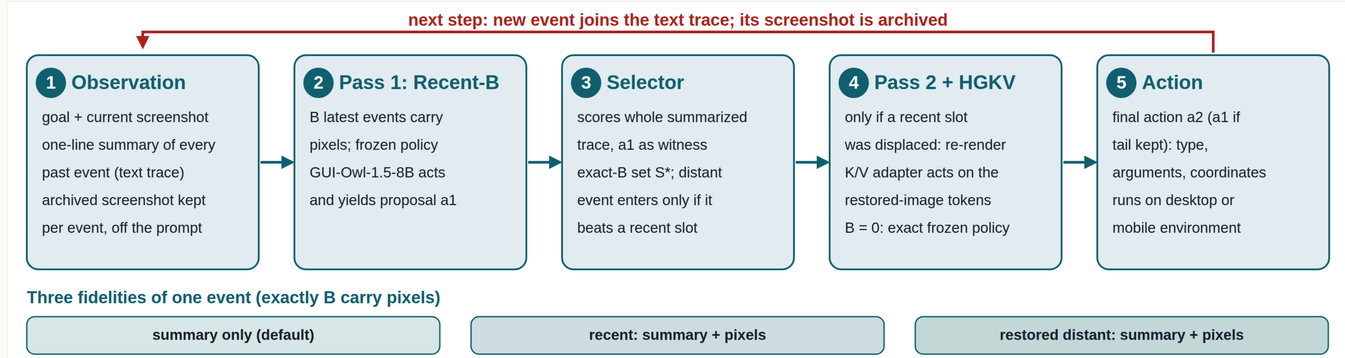
Our read (CUA Radar)

71	★★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; held-out mobile; 3 runs; matched budgets; cost; ablation. **Not reported:** code.
- **Flags:** none recorded.
- Pattern C15. Authors' numbers, not rerun.

Method: text summary for every event, pixels for B chosen ones

- **Sees / acts:** goal, text summaries, current screenshot and B restored images in; next GUI action out.
- **Mechanism:** a K/V adapter on restored-image tokens plus a selector that swaps in older events.
- **Signal:** matched-budget replacement groups, scored by the frozen policy's log-likelihood.



Our redraw (not the paper's figure): the agent loop with the learned selector, the same-budget re-decision and the history-gated adapter.

$$S^* = \arg \max_{S \subseteq \{1,\ldots,t-1\},\ |S| = \min(B, t-1)} Q(S)$$

Q: policy's mean log-likelihood of the successful next action

Key results: MobileWorld zero-shot success (%), B = 4

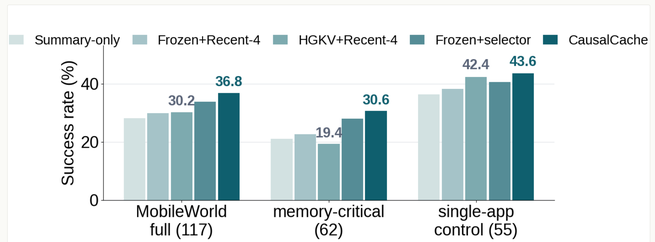


Table 2, redrawn: mean of three runs; 117 tasks = 62 memory-critical + 55 single-app controls.

Memory-critical 30.6 vs 19.4 (+11.2 pp); controls 43.6 vs 42.4

SO WHAT → *OSWorld, 15-step cap: 33.3 vs 33.0 (tie)*

Ablation: MobileWorld memory-critical success (%)

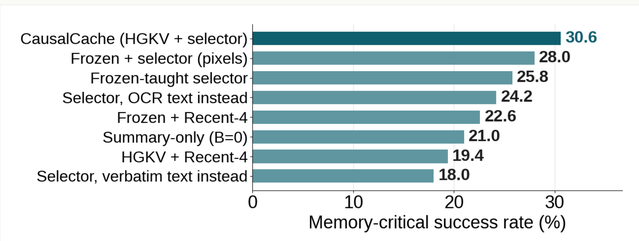


Table 2 and text, redrawn: 62 tasks, B = 4, online.

SO WHAT → *Selector on a frozen policy keeps most of the gain; text in place of restored pixels does not (OCR 24.2, verbatim 18.0 vs pixels 28.0). Same budget and prompt structure in every row; text-swap rows run on the frozen policy with the selector.*

Headline Numbers

36.8%
MobileWorld, zero-shot
vs 30.2% HGKV+Recent-4

- **30.6%** memory-critical
- **46.7%** OSWorld, 30 steps
- **33.3%** OSWorld, 15 steps

The paper's own controls

CONTROL	CAUSALCACHE	COMPARED WITH
MW single-app (55)	43.6	Recent-4 42.4
OSWorld 15 steps	33.3	Recent-4 33.0
OSWorld 30-step rerun	46.7	Recent-4 42.4
Offline: LL margin B=4	+.017	Random +.008

Cost: median 8 extra calls, ~7% of time

Takeaway

Choosing which past events regain pixels gives +11.2 pp on cross-app mobile tasks; OSWorld ties at 15 steps.

Frozen GUI-Owl-1.5-8B | k = 1 training | 30 steps: 46.7 vs 42.4

Scan to read the full paper

arxiv.org/abs/2608.22577v2